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COMPLETE LIST OF PUBLICATIONS AND RESEARCH OUTPUTS

COMPLETE LIST OF PUBLICATIONS

Authored Books



- [1] F. Magoulès, F.-X. Roux, and Q. Zou. 并行科算 (*Parallel Scientific Computing*). Publishing House of Electronics Industry, China, 2026. Hardcover 2012 pages (in Chinese).



- [2] F. Magoulès, H. Zhao, and C. Yimin. 建筑能耗分析中的数据挖掘与机器学习 (*Data Mining and Machine Learning in Building Energy Analysis*). China Machine Press, China, 2018. Paperback 153 pages (in Chinese).



- [3] F. Magoulès, F.-X. Roux, and P. Iványi. *Párhuzamos Számítási Módszerek (Parallel Scientific Computing)*. Pollack Press, Pécs, Hungary, 2018. Hardcover 196 pages (in Hungarian).



- [4] F. Magoulès and F.-X. Roux. *Calcul scientifique parallèle: Cours, exemples avec OpenMP et MPI, exercices corrigés, 2ème édition*. Collection Sciences Sup, Mathématiques / SMAI. Dunod, Paris, France, 2017. Paperback 286 pages (in French).



- [5] F. Magoulès and H.-X. Zhao. *Data Mining and Machine Learning in Building Energy Analysis*. Computer Engineering Series. Wiley-ISTE, London, UK, 2016. Hardcover 186 pages.



- [6] F. Magoulès, F.-X. Roux, and G. Houzeaux. *Parallel Scientific Computing*. Computer Engineering Series. Wiley-ISTE, London, UK, 2015. Hardcover 354 pages.



- [7] F. Magoulès, F.-X. Roux, and T. Kuwahara. *Heiretsu keisan-no suuri-to algorithm (Parallel Scientific Computing)*. Morikita Publishing Co Ltd, Tokyo, Japan, 2015. Paperback 246 pages (in Japanese).



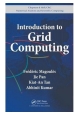
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- [10] F. Magoulès, J. Pan, and F. Teng. *Cloud-Computing: Data-Intensive Computing and Scheduling*. Chapman & Hall/CRC Numerical Analysis & Scientific Computing Series. CRC Press, Boca Raton, USA, 2012. Hardcover 231 pages.



- [11] F. Magoulès, J. Pan, K.-A. Tan, and A. Kumar. *Introduction to Grid Computing*. Chapman & Hall/CRC Numerical Analysis & Scientific Computing Series. CRC Press, Boca Raton, USA, 2009. Hardcover 334 pages.

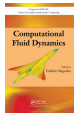


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Edited Books



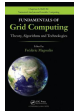
- [13] F. Magoulès, editor. *Patterns for Parallel Programming on GPUs*. Computational Science, Engineering and Technologies Series. Saxe-Coburg Publications, Stirlingshire, UK, 2014. Paperback 328 pages.



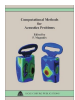
- [14] F. Magoulès, editor. *Computational Fluid Dynamics*. Chapman & Hall/CRC Numerical Analysis & Scientific Computing Series. CRC Press, Boca Raton, USA, 2011. Hardcover 407 pages.



- [15] F. Magoulès, editor. *Substructuring Techniques and Domain Decomposition Methods*. Saxe-Coburg Publications, Stirlingshire, UK, 2010. Hardcover 272 pages.



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- [17] F. Magoulès, editor. *Computational Methods for Acoustics Problems*. Saxe-Coburg Publications, Stirlingshire, UK, 2008. Hardcover 336 pages.



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- [19] F. Magoulès, editor. *Mesh Partitioning Techniques and Domain Decomposition Methods*. Saxe-Coburg Publications, Stirlingshire, UK, 2007. Hardcover 368 pages.



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- [21] P. Iványi, F. Magoulès, and B.H.V. Topping, editors. *Proceedings of the Seventh International Conference on Parallel, Distributed, GPU and Cloud Computing for Engineering*. Civil-Comp Press, Edinburgh, UK, 2023. Online volume: CCC 4.



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- [1] G. Gbikpi-Benissan and F. Magoulès. Asynchronous additive coarse-space correction. *SIAM Journal on Scientific Computing*, page (in press), 2026.
- [2] M. Chenaud, J. Alves, and F. Magoulès. How to learn physics: A practical review of scientific machine learning. *Archives of Computational Methods in Engineering*, page (in press), 2026.
- [3] Simian Gu, Yuxuan Wang, Shun Yao Zhu, Lei Yu, and F. Magoulès. A bayesian-optimization-based method for exploring the energy optimization potential of data centers. *Journal of Carbon Neutrality*, 1(2):97–126, 2026.
- [4] F. Magoulès, M. Menoux, and A. Rozanova-Pierrat. Frequency range non-Lipschitz parametric optimization of a noise absorption. *SIAM Journal on Control and Optimization*, 63(2):1267–1285, 2025.
- [5] X. Xiao, A.-R. Allouche, E. Dartois, F. Magoulès, and D. Peláez. Efficient exploration of high-dimensional configuration spaces for the generation of chemical datasets. *Journal of Chemical Information and Modeling*, 65(20):11094–11106, 2025.
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- [9] S. Alexandrov, M. Rynkovskaya, Y. Li, and F. Magoulès. Expansion of a hole in a thin plate considering arbitrary smooth pressure-independent yield criterion and work-hardening law. *Thin-Walled Structures*, 205(112518), 2024.

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